



UiO : **CEMO** – Centre for Educational Measurement
University of Oslo

Meta-Analyses in Education

From Theory to Practice with R



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TU Dortmund University 2025
February 2025

Workshop Content

Day 1, 20.02.2025

Time	Content	Teacher(s)
09:30-10:45	Setting the stage: ▪ Some housekeeping concerning this course ▪ Key steps in a meta-analysis and workflow ▪ Meta-analytic data sets ▪ Standard random-effects models	DC
10:45-11:00	<i>Second breakfast</i>	
11:00-12:15	Meta-analytic working models: ▪ Effect size multiplicity ▪ Random-effects models with robust variance estimation ▪ Moderator analyses via mixed-effects meta-regression	DC
12:15-13:00	<i>Lunch time</i>	
13:00-14:30	Meta-analytic working models: ▪ Multilevel meta-analysis	RS

Workshop Content

Day 2, 21.02.2025

Time	Content	Notes
09:30-10:45	Meta-analytic working models: ▪ Multilevel meta-analysis with correlational dependencies	RS
10:45-11:00	<i>Second breakfast</i>	
11:00-12:15	Publication bias ▪ Definition and forms of bias ▪ Overview of methods to assess and adjust for publication bias	RS
12:15-13:00	<i>Lunch time</i>	
13:00-14:30	Multivariate meta-analysis ▪ Overview of the approach Meta-analytic structural equation modeling (MASEM) ▪ Overview of the two MASEM approaches ▪ Example illustrating correlation-based MASEM	RS

Workshop Content

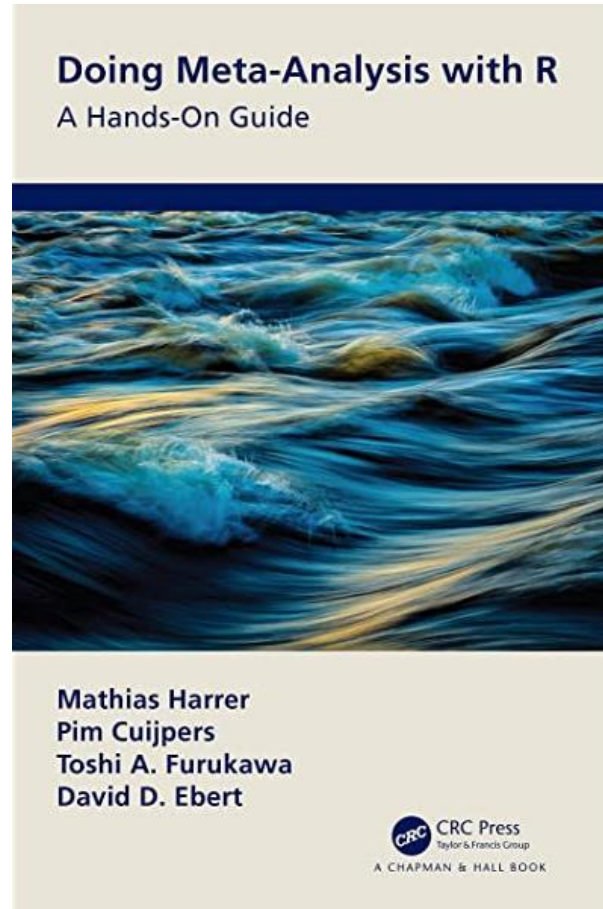
Basic concepts of synthesizing effect sizes

- Standard random-effects models
- Meta-analytic data structure
- Multilevel meta-analysis
- Moderator analyses
- Publication bias and influential effect sizes
- Multivariate meta-analysis

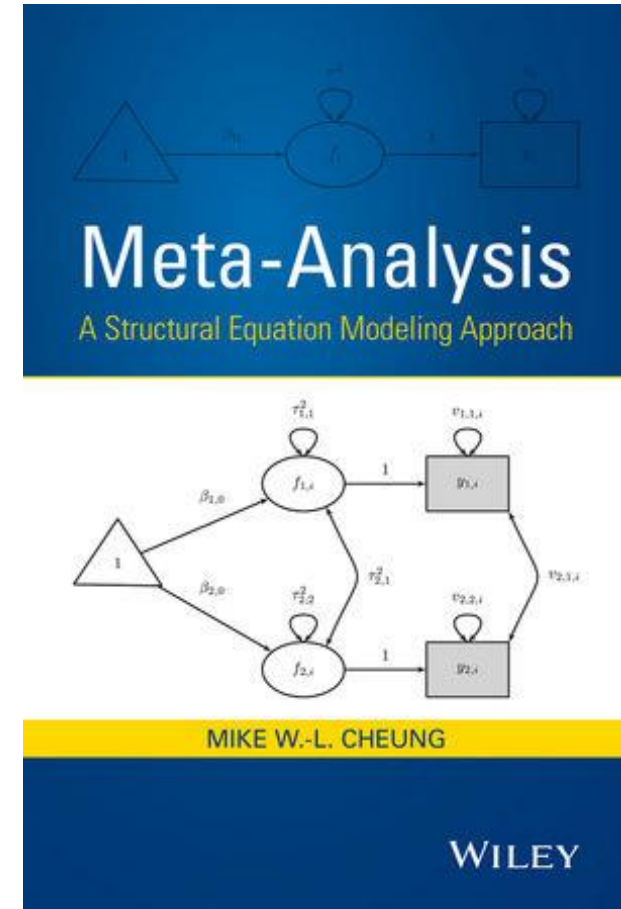
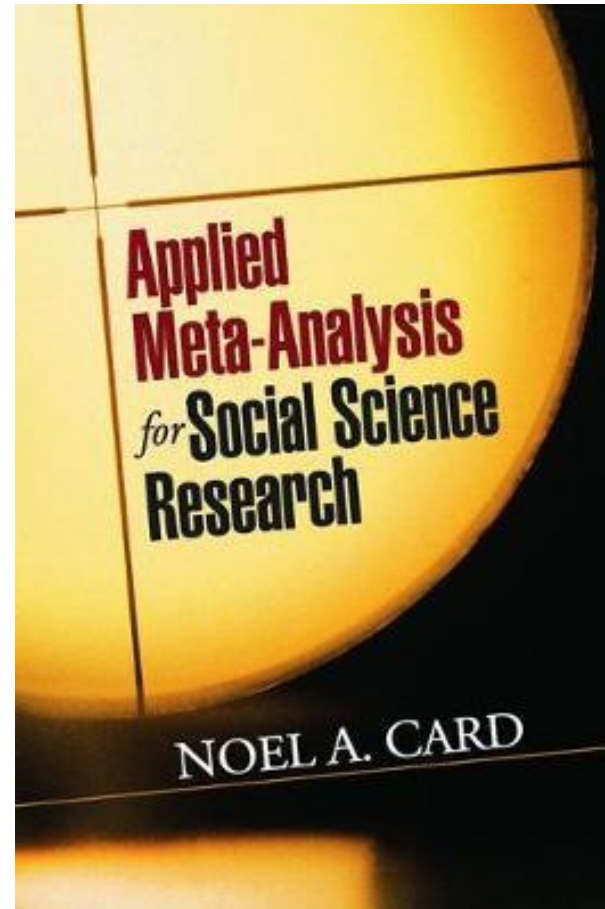
Data analyses and illustrative example using the R packages `metafor` and some supplementary packages



Recommended Literature



[https://bookdown.org/MathiasHarrer/
Doing_Meta_Analysis_in_R/](https://bookdown.org/MathiasHarrer/Doing_Meta_Analysis_in_R/)



[https://onlinelibrary.wiley.com/doi/
book/10.1002/9781118957813](https://onlinelibrary.wiley.com/doi/book/10.1002/9781118957813)

Steps in a Meta-Analysis

(Borenstein et al., 2009; Card, 2012)

1. Problem Formulation

3. Screening

5. Appraisal

7. Reporting



2. Literature Review

4. Data Extraction

6. Synthesis

8. Updating

Focus of this workshop:
Quantitative synthesis of effect sizes



What experience do you have with conducting meta-analyses?



BEGIN.

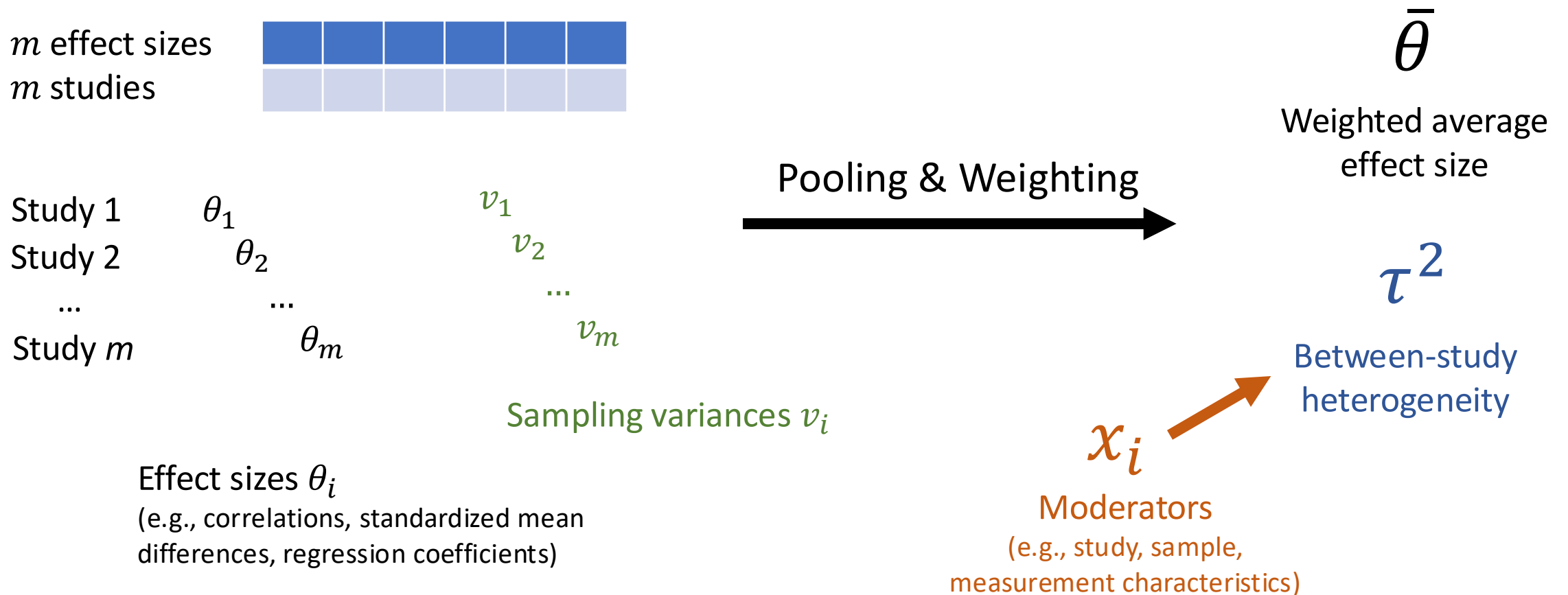
Main Purposes of Meta-Analyses

Three key outcomes of a meta-analysis—pooled effect size, heterogeneity, and moderator effects

Pooled Effect Size and Heterogeneity

(Borenstein et al., 2009)

Typical univariate meta-analysis



Variance Components in a Meta-Analysis

(CMM, 2019)

Two-level hierarchical structure

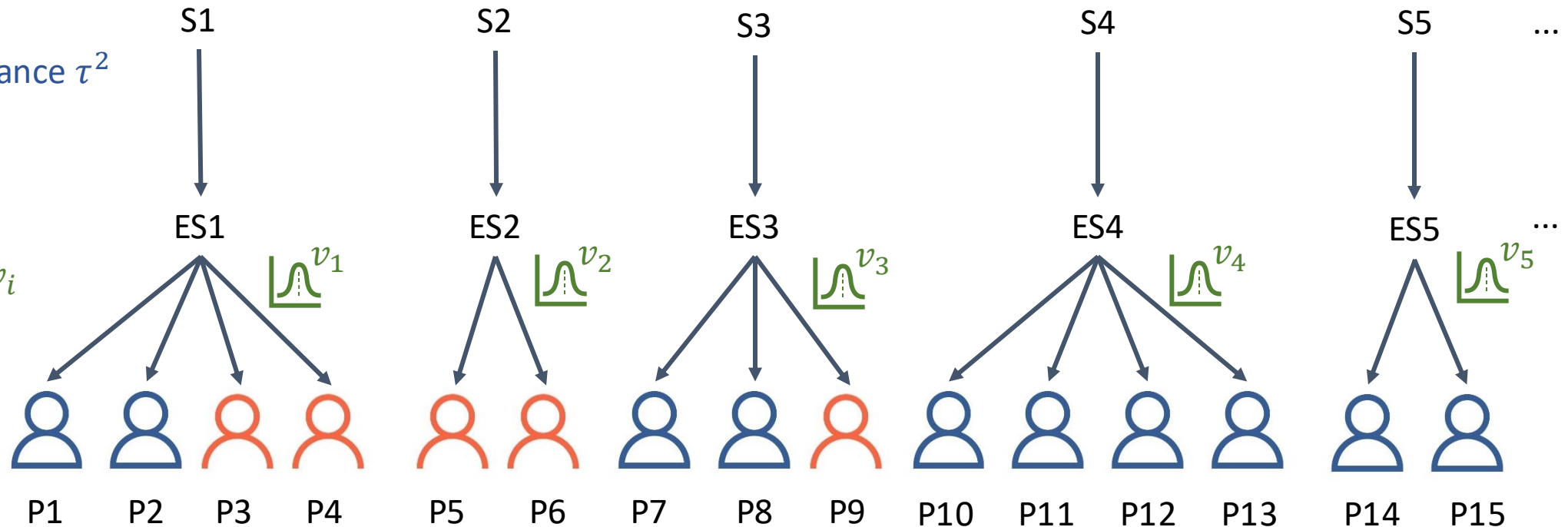
Studies (S)

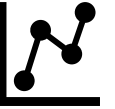
Between-study variance τ^2

Effect sizes (ES)

Sampling variance v_i

Participants (P)





Data Example: Gender Differences in Digital Literacy

(Campos et al., 2023; Scherer et al., 2024)

Meta-analytic data set of the gender differences in students' digital literacy

- Performance-based measures of digital literacy in K-12 student samples
- 59 effect sizes from 24 studies and 31 countries (both ILSA and non-ILSA)
- Standardized mean differences (female-male) Hedges' g (g) and sampling variance ($Var.g$)
- Identifiers of effect sizes ($ESID$), primary studies ($IDSTUDY$), and countries ($COUNTRY$)
- Sample sizes (N , n_F , n_M)
- Contextual variables: Publication year ($PubYear$), availability of individual participant data (IPD), countries' power distance index (PDI), and GDP ($cGCP$)

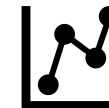
Research questions:

1. To what extent do boys and girls differ in their digital literacy performance?
2. To what extent do the gender effects vary across studies and countries?
3. Which contextual variables explain the possible heterogeneity in the effects?

Pooled effect size

Heterogeneity

Moderators



Data Example: Gender Differences in Digital Literacy

(Campos et al., 2023; Scherer et al., 2024)

Importing and exploring data in R

```
# Read data
dat <- read_excel("Gender_data_simp.xlsx")

# Data structure
str(dat)
```

```
## tibble [59 × 17] (S3: tbl_df/tbl/data.frame)
## $ ESID      : num [1:59] 1 2 3 4 5 6 7 8 9 10 ...
## $ Reference : chr [1:59] "ACARA (2012a,b)" "ACARA (2012a,b)" "ACER (2008)" "ACER (2008)" ...
## $ IDSTUDY   : num [1:59] 1 1 2 2 3 4 5 6 7 8 ...
## $ Sample name: chr [1:59] "ACARA (2012a,b)-Grade 6" "ACARA (2012a,b)-Grade 10" "ACER (2008)-Grade 6"
## $ IDSAMPLE  : num [1:59] 1 2 3 4 5 6 7 8 9 10 ...
## $ Country   : chr [1:59] "Australia" "Australia" "Australia" "Australia" ...
## $ IDCOUNTRY : num [1:59] 36 36 36 36 56 410 380 840 344 36 ...
## $ PubYear   : num [1:59] 2012 2012 2008 2008 2015 ...
## $ N         : num [1:59] 5710 5313 5604 5322 378 ...
## $ nF        : num [1:59] 2741 2444 2752 2650 189 ...
## $ nM        : num [1:59] 2969 2869 2852 2672 189 ...
## $ g         : num [1:59] 0.1104 0.0667 0.0862 0.0744 0.3766 ...
## $ Var.g     : num [1:59] 0.0007 0.000758 0.000715 0.000752 0.010723 ...
## $ SE.g      : num [1:59] 0.0265 0.0275 0.0267 0.0274 0.1036 ...
## $ IPD       : num [1:59] 0 0 0 0 0 0 0 0 0 1 ...
## $ PDI       : num [1:59] 38 38 38 38 65 60 50 40 68 38 ...
## $ cGDP      : num [1:59] 57398 57398 57398 57398 40992 ...
```


Data Example: Gender Differences in Digital Literacy

Importing and exploring data in R

This data set contains several identifiers to indicate the complex meta-analytic data structure:

- **ESID:** Identifier of the effect size
- **IDSTUDY:** Identifier of the primary study
- **IDSAMPLE:** Identifier of the study sample
- **Country:** Identifier of the country

```
## Primary studies
table(dat$IDSTUDY)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
##  2  2  1  1  1  1  1 21 13  1  1  1  1  1  1  1  1  1  2  1  1  1  1
```

```
## Study samples
table(dat$IDSAMPLE)
```

[illegible]

```
## Countries
table(dat$Country)
```

##				
##	Argentina	Australia	Belgium (Flanders)	Canada
##	1	7	1	2
##	Chile	China	Croatia	Czech Republic
##	2	1	1	1
##	Denmark	Finland	France	Germany
##	2	1	1	4
##	Hong Kong SAR	Hungary	Italy	Kazakhstan
##	2	1	2	1
##	Korea, Republic of	Lithuania	Luxembourg	Norway
##	4	1	1	6
##	Poland	Portugal	Russian Federation	Slovak Republic
##	1	1	2	1
##	Slovenia	Switzerland	Thailand	The Netherlands
##	1	1	1	3
##	Turkey	Uruguay	USA	
##	1	1	4	

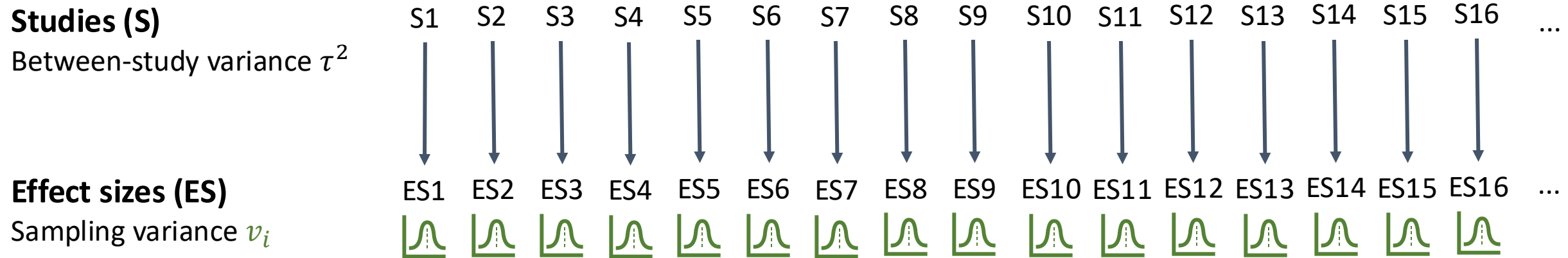
Standard Random-Effects Model

Meta-analytic baseline model assuming heterogeneity between effect sizes

Standard Meta-Analytic Data Structure

(CMM, 2019)

Two-level hierarchical structure: the «ideal» scenario



Level 1 is the level containing the **sampling variances** for each study (ES).

Level 2 is the level containing the **heterogeneity between studies** (S).

Standard Random-Effects Model

(Card, 2012; Cheung, 2015; Pastor & Lazowski, 2018)

Univariate random-effects model (REM)

for each study $i = 1, \dots, m$

Level 1:

$$\theta_i = f_i + e_i$$
$$e_i \sim N(0, v_i)$$

Level 2:

$$f_i = \beta_R + u_i$$
$$u_i \sim N(0, \tau^2)$$

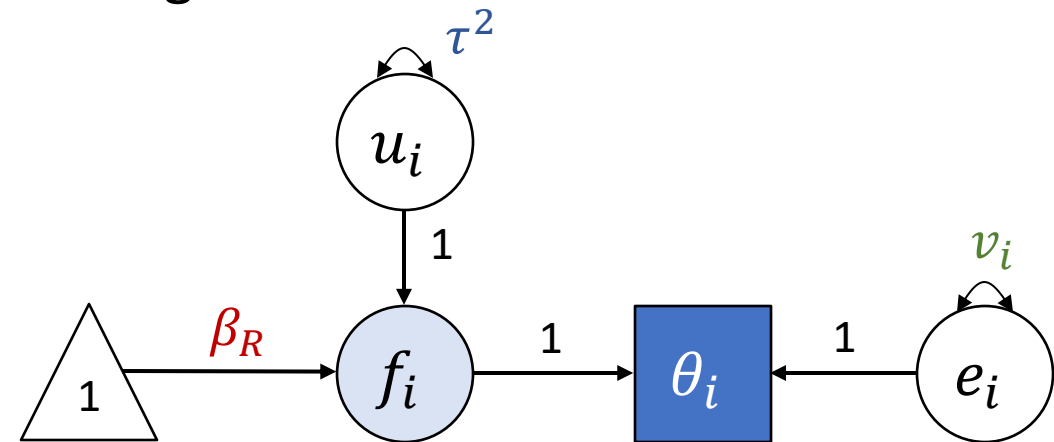
Total:

$$\theta_i = \beta_R + u_i + e_i$$

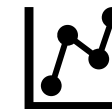
Weights:

$$w_i^* = 1/(\tau^2 + v_i)$$

Path diagram:



β_R : Weighted average effect size
under the REM



Standard Random-Effects Model

Univariate REM

Model estimation in metafor

```
## Model estimation
REM <- rma.mv(g,
              Var.g,
              random = list(~ 1 | ESID),
              method = "REML",
              data = dat)
```

```
## Model summary
summary(REM, digits = 4)
```

Alternative specification:

```
REM2 <- rma(g, Var.g, data = dat, method = "REML")
summary(REM2)
```

```
## Multivariate Meta-Analysis Model (k = 59; method: REML)
```

```
##
```

	logLik	Deviance	AIC	BIC	AICc
	26.7148	-53.4296	-49.4296	-45.3088	-49.2115

```
##
```

Variance Components: $\tau^2 = 0.0156$ heterogeneity

```
##
```

	estim	sqrt	nlvls	fixed	factor
sigma^2	0.0156	0.1247	59	no	ESID

```
##
```

Test for Heterogeneity:

Q(df = 58) = 592.4533, p-val < .0001

```
##
```

Model Results: $\bar{\theta} = 0.1456$ pooled effect size

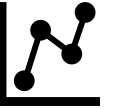
```
##
```

	estimate	se	zval	pval	ci.lb	ci.ub
	0.1456	0.0178	8.1906	<.0001	0.1107	0.1804 ***

```
##
```

```
## ---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



Testing for Heterogeneity

(Higgins & Thomson, 2002; Sánchez-Meca & Marín-Martínez, 2008)

I^2 statistic

- The proportion of observed variation reflecting true variation between effect sizes.
- It only indicates what proportion of the variation between effect sizes is true variation.
- 25% small, 50% medium, 75% large heterogeneity

```
## Number of effect sizes
k <- REM$k
## Weights from the model
REM.wi <- 1/REM$vi
REM.vt <- (k-1) * sum(REM.wi) / (sum(REM.wi)^2 - sum(REM.wi^2))
100 * REM$sigma2 / (REM$sigma2 + REM.vt)
```

```
## [1] 91.72972
```

$I^2 = 91.7\%$ large amount of heterogeneity

Confidence interval CI of the variance indicating the heterogeneity between effect sizes τ^2

95% CI of τ^2

```
## Variances: 95 % confidence intervals
confint(REM, digits = 4)
```

##	estimate	ci.lb	ci.ub
## sigma^2	0.0156	0.0099	0.0253
## sigma	0.1247	0.0993	0.1591

Zero is not included.

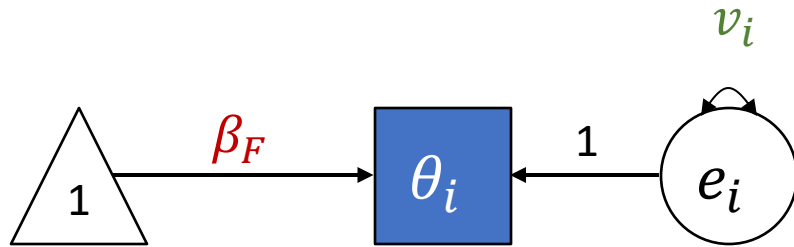
Testing for Heterogeneity

(Cheung, 2015; Hedges & Vevea 1998)

Model comparison

Fixed-effects model ($\tau^2 = 0$) vs. random-effects model (τ^2 freely estimated)

FEM

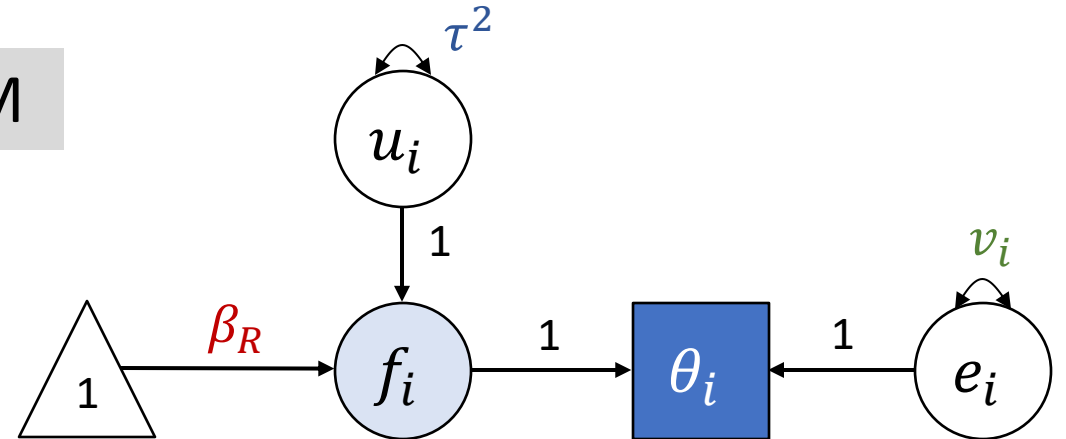


$$\theta_i = \beta_F + e_i$$
$$e_i \sim N(0, v_i)$$

Weights:

$$w_i^* = 1/v_i$$

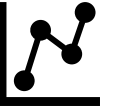
REM



$$\theta_i = \beta_R + u_i + e_i$$
$$e_i \sim N(0, v_i), u_i \sim N(0, \tau^2)$$

Weights:

$$w_i^* = 1/(\tau^2 + v_i)$$



Testing for Heterogeneity

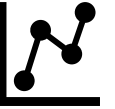
Model comparison

Random-effects model (τ^2 freely estimated) vs. fixed-effects model ($\tau^2 = 0$)

Estimate the FEM (in addition to the REM)

```
FEM <- rma.mv(g,
  Var.g,
  random = list(~ 1 | ESID),
  sigma2 = 0, # No heterogeneity
  method = "REML",
  data = dat)
```

```
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##      logLik      Deviance      AIC      BIC      AICc
## -170.9519    341.9037    343.9037    345.9642    343.9752
##
## Variance Components:   $\tau^2 = 0$  no heterogeneity
##
##              estim      sqrt  nlvls  fixed  factor
## sigma^2      0.0000    0.0000     59    yes    ESID
##
## Test for Heterogeneity:
## Q(df = 58) = 592.4533, p-val < .0001
##
## Model Results:
##
## estimate      se      zval      pval      ci.lb      ci.ub
##    0.1535    0.0049   31.6424   <.0001    0.1440    0.1630   ***
```



Testing for Heterogeneity

Model comparison

Random-effects model (τ^2 freely estimated) vs. fixed-effects model ($\tau^2 = 0$)

Perform a likelihood-ratio test and compare the information criteria

```
## Model comparison: REM vs. FEM  
anova(REM, FEM)
```

##	df	AIC	BIC	AICc	logLik	LRT	pval	QE
## Full	2	-49.4296	-45.3088	-49.2115	26.7148			592.4533
## Reduced	1	343.9037	345.9642	343.9752	-170.9519	395.3334	<.0001	592.4533

```
metafor::fitstats.rma(REM, FEM)
```

##	REM	FEM
## logLik:	26.71482	-170.9519
## deviance:	-53.42964	341.9037
## AIC:	-49.42964	343.9037
## BIC:	-45.30875	345.9642
## AICc:	-49.21146	343.9752

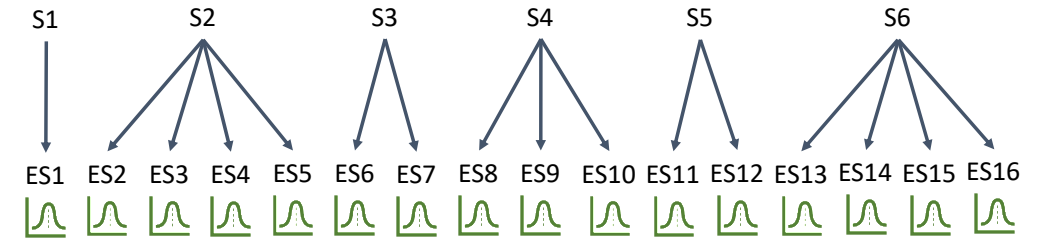
Result: The REM is preferred over the FEM. This is evidence for heterogeneity between effect sizes.

Effect Size Multiplicity

Multiple effects sizes available per study

Effect Size Multiplicity—Why?

Sources of multiple effect sizes per study (“Effect size multiplicity”)



- Multiple **populations or sub-populations** (e.g., different samples or age groups)
- Multiple **treatment or control groups** (e.g., multiple intervention arms, active/passive controls)
- Multiple **outcome variables** (e.g., multiple constructs, different ways of measuring the same construct)
- Multiple **time points** (e.g., multiple post-tests)
- ...

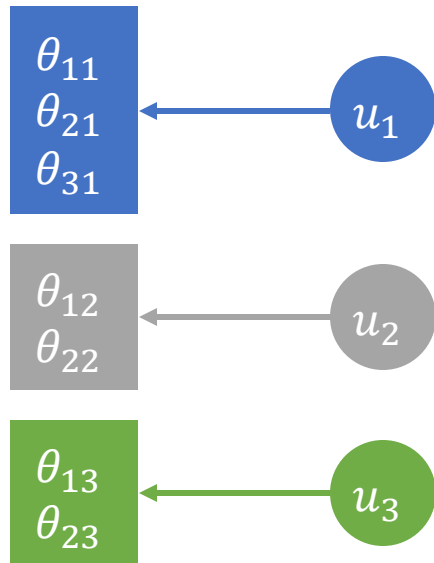
...or combinations 😊

(López-López et al., 2018, <https://doi.org/10.1002/jrsm.1310>)

Types of Effect Size Multiplicity

(Pustejovsky & Tipton, 2021, <https://doi.org/10.1007/s11121-021-01246-3>;
Illustration inspired by J. E. Pustejovsky, UseR! Oslo Talk, 02.09.2021)

Correlated Effects.



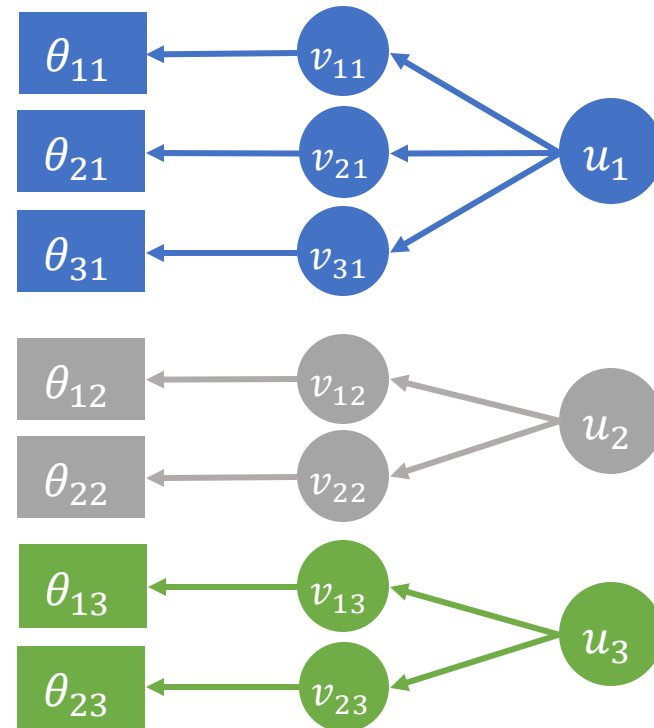
Correlated effect sizes
Between-study heterogeneity
No within-study heterogeneity

Examples: Multiple measures of the same construct, multiple measurement occasions of the same sample

Strategy

Specify the correlation between effect sizes

Hierarchical Effects.



Effect sizes not correlated
Between-study heterogeneity
Within-study heterogeneity

Examples: Multiple samples included in a study, multi-lab replications

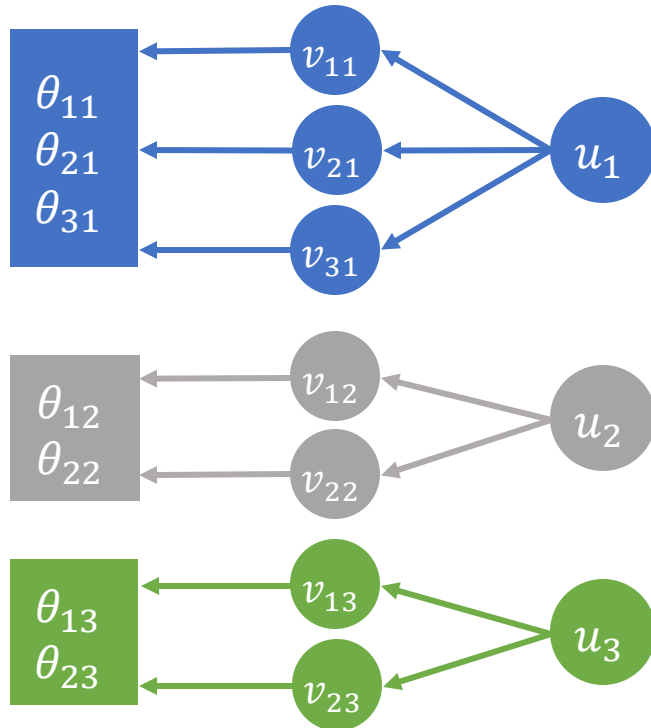
Strategy

Estimate variances at different levels

Types of Effect Size Multiplicity

(Pustejovsky & Tipton, 2021, <https://doi.org/10.1007/s11121-021-01246-3>;
Illustration inspired by J. E. Pustejovsky, UseR! Oslo Talk, 02.09.2021)

Correlated *and* Hierarchical Effects.



Correlated effect sizes
Between-study heterogeneity
Within-study heterogeneity

Example: Multiple measures available from multiple samples within a study

Strategy

Estimate variances at different levels &
Specify the correlation between effect sizes

Effect Size Multiplicity

Meta-analytic data set of the gender differences in students' digital literacy

- Performance-based measures of digital literacy in K-12 student samples
- 59 effect sizes from 24 studies and 31 countries (both ILSA and non-ILSA)
- Standardized mean differences (female-male) Hedges' g (g) and sampling variance ($\text{Var. } g$)
- Identifiers of effect sizes (ESID), primary studies (IDSTUDY), and countries (IDCOUNTRY)

```
## Primary studies
table(dat$IDSTUDY)
```

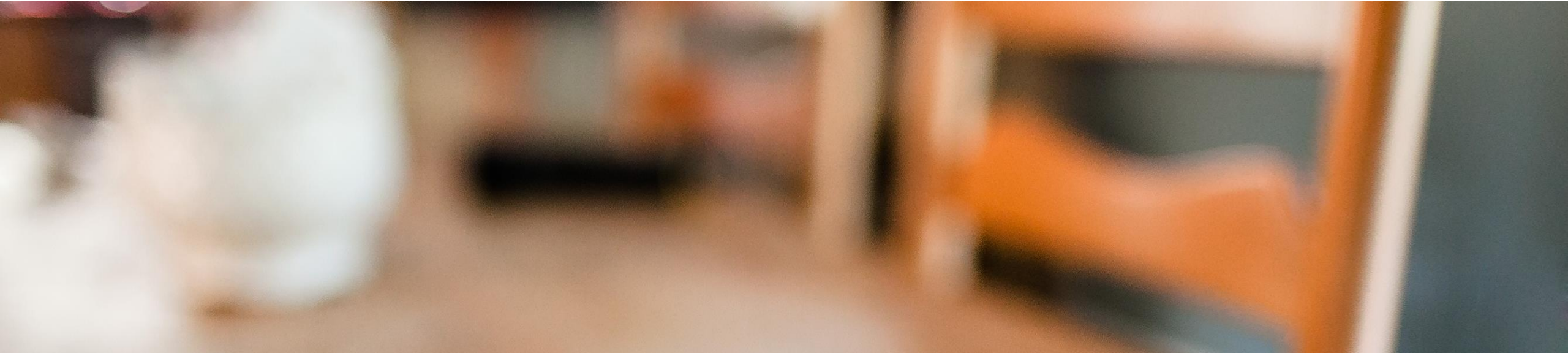
```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
##  2  2  1  1  1  1  1 21 13  1  1  1  1  1  1  1  1  1  2  1  1  1  1
```

```
## Study samples
table(dat$IDSAMPLE)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
##  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
##  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## 53 54 55 56 57 58 59
##  1  1  1  1  1  1  1
```

```
## Countries
table(dat$Country)
```

##				
##	Argentina	Australia	Belgium (Flanders)	Canada
##	1	7	1	2
##	Chile	China	Croatia	Czech Republic
##	2	1	1	1
##	Denmark	Finland	France	Germany
##	2	1	1	4
##	Hong Kong SAR	Hungary	Italy	Kazakhstan
##	2	1	2	1
##	Korea, Republic of	Lithuania	Luxembourg	Norway
##	4	1	1	6
##	Poland	Portugal	Russian Federation	Slovak Republic
##	1	1	2	1
##	Slovenia	Switzerland	Thailand	The Netherlands
##	1	1	1	3
##	Turkey	Uruguay	USA	
##	1	1	4	



What type of effect size multiplicity might you encounter when conducting a meta-analysis in your research field?

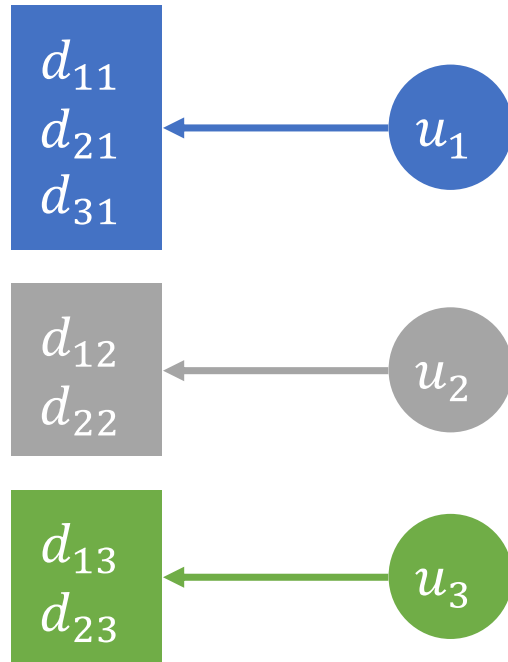


BEGIN.

Robust Variance Estimation (RVE)

Correct standard errors and test statistics for effect size multiplicity

Meta-Analysis with Robust Variance Estimation



Correlated Effects.

The only source of heterogeneity is at the *study* level

- *Within* a study, all effect sizes measure the *same* underlying population effect

$$\theta_{ij} = \beta_R + u_j + e_{ij}$$

$$Var(u_j) = \tau^2, Var(e_{ij}) = s_j^2, Cov(e_{hj}, e_{ij}) = \rho s_j^2$$

Weights for each effect size i in study j

$$w_{ij} = \frac{1}{n_j(\hat{\tau}^2 + s_j^2)}$$

Account for an overall correlation between effects (not: study-specific)

Meta-Analysis with Robust Variance Estimation

RVE assuming correlated effects

```
#RVE assuming correlated effects
```

```
RVEce.study <- robu(g ~ 1,  
  data = dat,  
  studynum = IDCOUNTRY,  
  var.eff.size = Var.g,  
  small = FALSE,  
  modelweights = "CORR",  
  rho = 0.5  
)
```

```
## RVE: Correlated Effects Model
```

```
##
```

```
## Model: g ~ 1
```

```
##
```

```
## Number of studies = 31
```

```
## Number of outcomes = 59 (min = 1 , mean = 1.9 , median = 1 , max = 7 )
```

```
## Rho = 0.5
```

```
## I.sq = 86.03989
```

```
## Tau.sq = 0.01113374
```

```
##
```

##		Estimate	StdErr	t-value	dfs	P(t >)	95% CI.L	95% CI.U	Sig
##	1 X.Intercept.	0.139	0.0186	7.51	30	0.0000000226	0.101	0.177	***

```
## ---
```

```
## Signif. codes: < .01 *** < .05 ** < .10 *
```

```
## ---
```

β_R

Meta-Analysis with Robust Variance Estimation

Sensitivity analyses are mandatory to determine the effects of different correlations among multiple effect sizes

```
## Sensitivity Analyses
sensitivity(RVEce.study)
```

```
## RVE: Correlated Effects Model
## Model: g ~ 1
##
## Sensitivity Analysis
##
##
```

		Rho = 0	Rho = 0.2	Rho = 0.4	Rho = 0.6	Rho = 0.8
##	X.Intercept. Coefficient	0.1394	0.1394	0.1394	0.1394	0.1394
##	Std. Error	0.0186	0.0186	0.0186	0.0186	0.0186
##	Tau.sq Estimate	0.0111	0.0111	0.0111	0.0111	0.0111

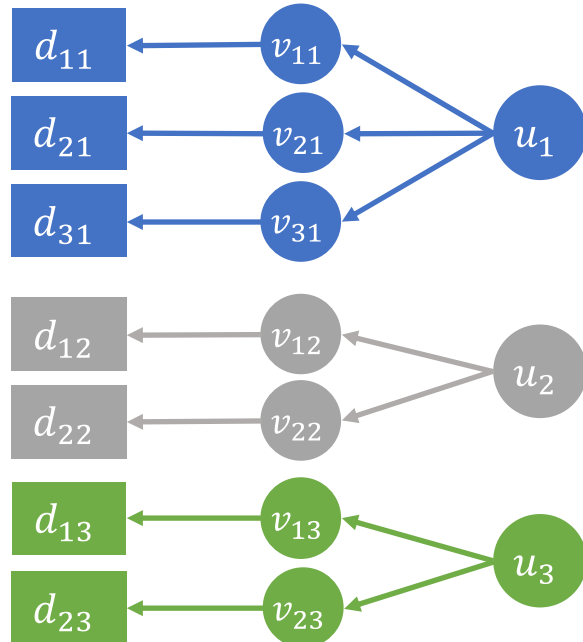
```
## Rho = 1
## 0.1394
## 0.0186
## 0.0111
```

Meta-Analysis with Robust Variance Estimation

Hierarchical Effects.

The model posits two separate levels of random variation

- **Between-Study variation** and **Within-study variation**

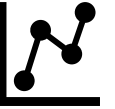


$$\theta_{ij} = \beta_R + u_j + v_{ij} + e_{ij}$$

$$Var(u_j) = \tau^2, Var(v_{ij}) = \omega^2, Var(e_{ij}) = s_{ij}^2, Cov(e_{hj}, e_{ij}) = 0$$

Weights for each effect size i in study j : $w_{ij} = 1/(\tau^2 + \omega^2 + s_{ij}^2)$

This is a multilevel meta-analytic model, but between- and within-study heterogeneity are incidental.



Meta-Analysis with Robust Variance Estimation

Random-effects model with RVE

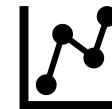
Effect sizes nested in countries

```
## Clustering: Countries
RVE.count <- robu(g ~ 1,
                  data = dat,
                  studynum = IDCOUNTRY,
                  var.eff.size = Var.g,
                  small = FALSE,
                  modelweights = "HIER")
```

```
## Model summary
print(RVE.count)
```

```
## RVE: Hierarchical Effects Model
##
## Model: g ~ 1
##
## Number of clusters = 31
## Number of outcomes = 59 (min = 1 , mean = 1.9 , median = 1 , max = 7 )
## Omega.sq = 0.01131715
## Tau.sq = 0.001700087
##
##      Estimate StdErr t-value dfs      P(|t|>) 95% CI.L 95% CI.U Sig
## 1 X.Intercept.    0.146 0.0186   7.85  30 0.00000000936    0.108    0.184 ***
## ---
## Signif. codes: < .01 *** < .05 ** < .10 *
## ---
```

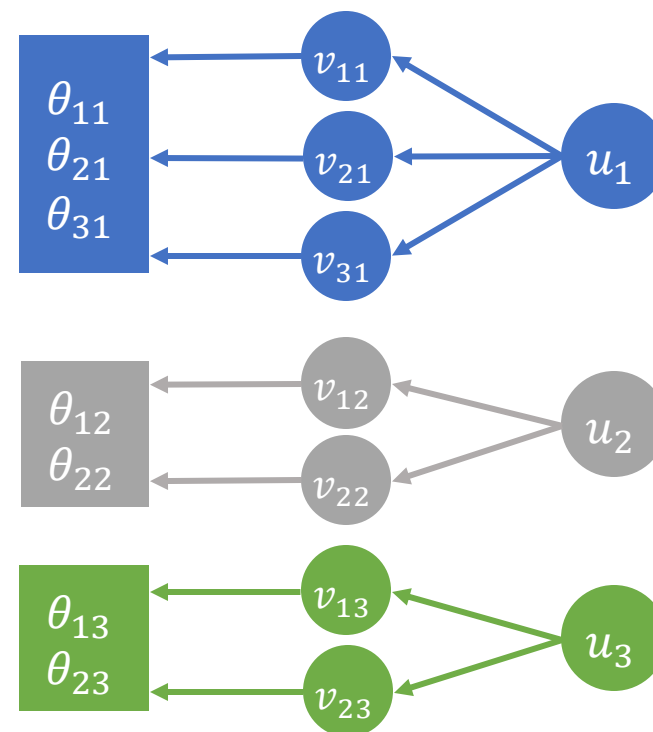
β_R



Meta-Analysis with Robust Variance Estimation

Steps in a Typical Analysis:

1. Select a dependency structure
2. Estimate variance components
3. Weight effect sizes
4. Calculate robust standard errors



Moderator Analyses

Mixed-effects meta-regression models to explain heterogeneity

Moderator Analysis

(Cheung, 2015)

Two-level mixed-effects meta-regression

i : Effect sizes, j : Studies

Level 1:

$$\theta_i = f_i + e_i$$

$e_i \sim N(0, v_i)$

Level 2:

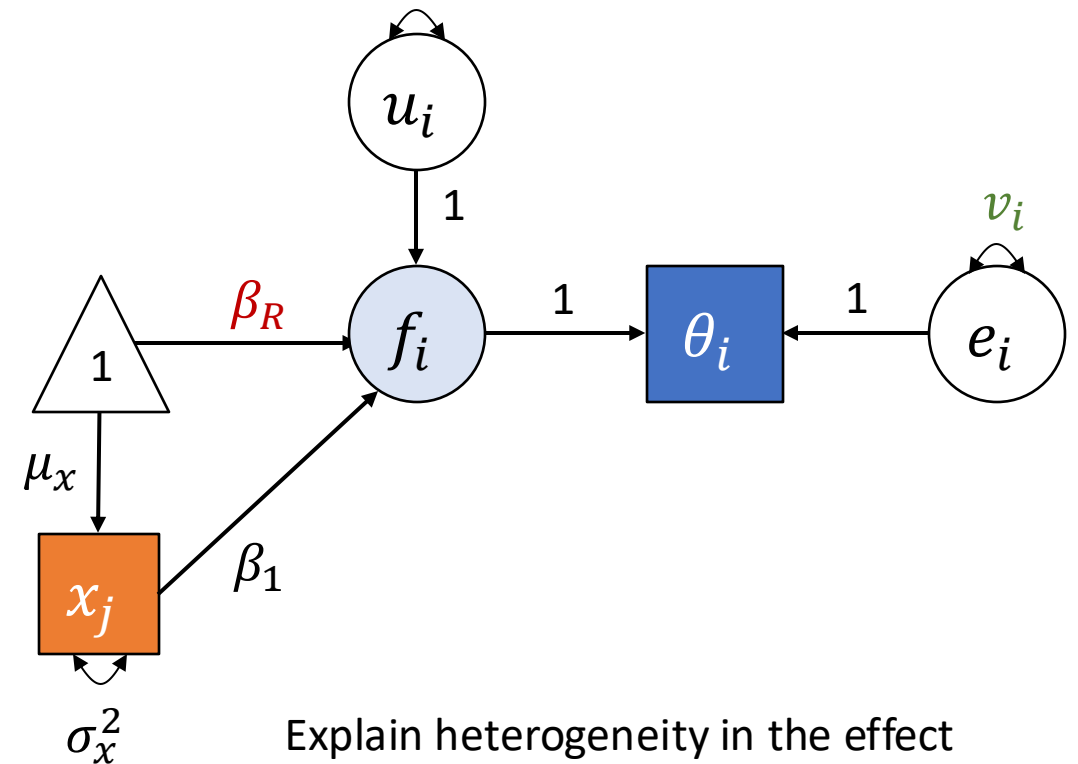
$$f_i = \beta_R + \beta_1 x_j + u_i$$

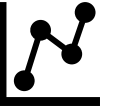
$u_i \sim N(0, \tau^2)$

Total:

$$\theta_i = \beta_R + \beta_1 x_j + u_i + e_i$$

Path diagram:





Moderator Analysis

Mixed-effects meta-regression

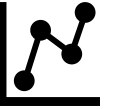
Moderation by the availability of IPD (binary variable)

```
## Mixed-effects meta-regression model
REM.ipd <- rma.mv(g,
  Var.g,
  random = list(~ 1 | ESID),
  method = "REML",
  data = dat,
  tdist = TRUE,
  test = "t",
  mods =~ IPD)
```

Result: No evidence on the differences in effect sizes between studies with and without IPD.

```
##
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##   logLik  Deviance      AIC      BIC      AICc
## 26.4207  -52.8414  -46.8414  -40.7122  -46.3886
##
## Variance Components:
##
##           estim      sqrt  nlvls  fixed  factor
## sigma^2    0.0153  0.1236    59     no    ESID
##
## Test for Residual Heterogeneity:
## QE(df = 57) = 578.8574, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 57) = 1.5286, p-val = 0.2214
##
## Model Results:
##
##           estimate      se    tval  df    pval    ci.lb    ci.ub
## intrcpt    0.1170  0.0291  4.0192  57  0.0002  0.0587  0.1752 ***
## IPD        0.0452  0.0366  1.2364  57  0.2214 -0.0280  0.1185
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

β_1



Moderator Analysis

Mixed-effects meta-regression

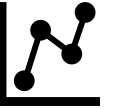
Moderation by the availability of IPD (binary variable)

Model estimation without the intercept

```
REM.ipd2 <- rma.mv(g,
  Var.g,
  random = list(~ 1 | ESID),
  method = "REML",
  data = dat,
  tdist = TRUE,
  test = "t",
  mods =~ factor(IPD) - 1 # remove the intercept
)
```

This way, the weighted average effect sizes for each group are estimated directly (assuming the same amount of heterogeneity for each group).

```
##
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##   logLik  Deviance      AIC      BIC      AICc
##  26.4207  -52.8414  -46.8414  -40.7122  -46.3886
##
## Variance Components:
##
##           estim  sqrt  nlvls  fixed  factor
## sigma^2    0.0153  0.1236    59    no    ESID
##
## Test for Residual Heterogeneity:
## QE(df = 57) = 578.8574, p-val < .0001
##
## Test of Moderators (coefficients 1:2):
## F(df1 = 2, df2 = 57) = 34.8521, p-val < .0001
##
## Model Results:
##
##           estimate      se    tval  df    pval    ci.lb    ci.ub
## factor(IPD)0    0.1170  0.0291  4.0192  57  0.0002  0.0587  0.1752 ***
## factor(IPD)1    0.1622  0.0222  7.3178  57 <.0001  0.1178  0.2066 ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Moderator Analysis

Mixed-effects meta-regression

Moderation by the Publication Year (continuous variable)

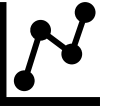
```
## Mixed-effects meta-regression model
REM.py <- rma.mv(g,
  Var.g,
  random = list(~ 1 | ESID),
  method = "REML",
  data = dat,
  tdist = TRUE,
  test = "t",
  mods =~ scale(PubYear, ## standardized
    center = TRUE,
    scale = TRUE))

# Summarize the results
summary(REM.py, digits=4)
```

Result: There is evidence on the relation between the effects sizes and publication year.

```
##
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##      logLik  Deviance      AIC      BIC      AICc
## 31.3326 -62.6651 -56.6651 -50.5360 -56.2123
##
## Variance Components:
##
##              estim  sqrt  nlvls  fixed  factor
## sigma^2      0.0123  0.1108    59    no    ESID
##
## Test for Residual Heterogeneity:
## QE(df = 57) = 470.8095, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 57) = 12.6320, p-val = 0.0008
##
## Model Results:
##
##              estimate      se    tval  df
## intrcpt          0.1426  0.0161  8.8846  57
## scale(PubYear, center = TRUE, scale = TRUE) 0.0566  0.0159  3.5541  57
##
##              pval    ci.lb    ci.ub
## intrcpt          <.0001  0.1105  0.1748 ***
## scale(PubYear, center = TRUE, scale = TRUE) 0.0008  0.0247  0.0885 ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

β_1



Moderator Analysis

Mixed-effects meta-regression

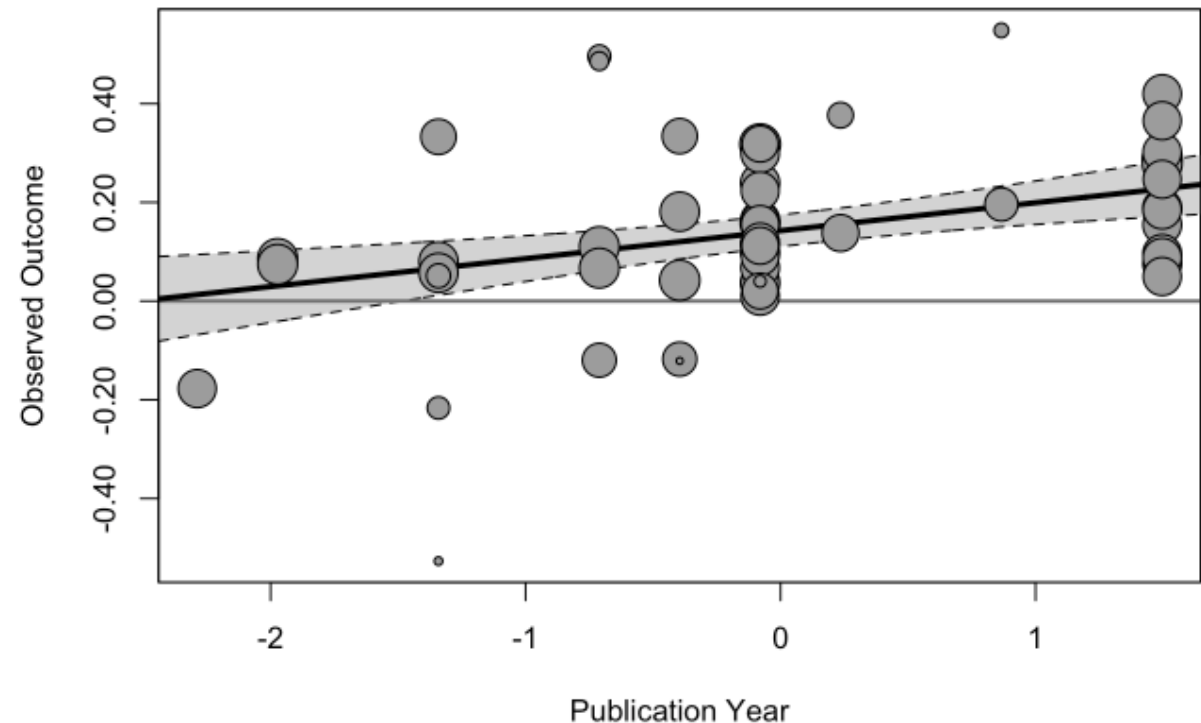
Moderation by the PDI (continuous variable)

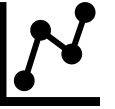
Level-specific variance explanation as the proportional reduction of variance (R^2)

```
## Level: Primary studies  
max(1-REM.py$sigma2[1]/REM$sigma2[1] , 0)
```

```
## [1] 0.2111067
```

```
## Regression plot  
regplot(REM.py,  
        xlab = "Publication Year",  
        refline = 0,  
        digits = 2)
```





RVE: Moderator Analysis

Mixed-effects meta-regression

Moderation by the Publication Year (continuous variable)

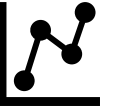
RVE with Hierarchical Effects

```
## RVE Model with hierarchical effects and Publication Year
RVE.count.py <- robu(g ~ 1 + scale(PubYear, center = TRUE, scale = TRUE),
  data = dat,
  studynum = IDCOUNTRY,
  var.eff.size = Var.g,
  small = FALSE,
  modelweights = "HIER")
```

Result: There is evidence on the relation
between the effects sizes and publication year.

```
## RVE: Hierarchical Effects Model
##
## Model: g ~ 1 + scale(PubYear, center = TRUE, scale = TRUE)
##
## Number of clusters = 31
## Number of outcomes = 59 (min = 1 , mean = 1.9 , median = 1 , max = 7 )
## Omega.sq = 0.00992801
## Tau.sq = 0.0003710178
##
##
##                                     Estimate StdErr t-value dfs
## 1                                     X.Intercept.  0.14247 0.01647   8.652  29
## 2 scale.PubYear..center...TRUE..scale...TRUE.  0.05608 0.01626   3.450  29
##
##          P(|t|>) 95% CI.L 95% CI.U Sig
## 1 0.0000000001583  0.10879  0.17615 ***
## 2 0.001736958944  0.02284  0.08933 ***
## ---
## Signif. codes: < .01 *** < .05 ** < .10 *
## ---
```

β_1



Moderator Analysis

Mixed-effects meta-regression

Moderation by Power Distance and Country GDP

```
## Mixed-effects meta-regression model
REM.pgd <- rma.mv(g,
  Var.g,
  random = list(~ 1 | ESID),
  method = "REML",
  data = dat,
  tdist = TRUE,
  test = "t",
  mods =~ scale(PDI, ## standardized
    center = TRUE,
    scale = TRUE) +
    scale(cGDP, ## standardized
    center = TRUE,
    scale = TRUE))

# Summarize the results
summary(REM.pgd, digits=4)
```

```
##
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##   logLik  Deviance      AIC      BIC      AICc
## 25.6126  -51.2251  -43.2251  -35.1237  -42.4408
##
```

Variance Components:

```
##
##           estim      sqrt  nlvls  fixed  factor
## sigma^2    0.0159  0.1261    59    no    ESID
##
```

Test for Residual Heterogeneity:

```
## QE(df = 56) = 589.6128, p-val < .0001
##
```

Test of Moderators (coefficients 2:3):

```
## F(df1 = 2, df2 = 56) = 1.0476, p-val = 0.3576
##
```

Model Results:

```
##
##                                     estimate      se      tval  df      pval
## intrcpt                           0.1453  0.0179  8.0962  56  <.0001
 $\beta_1$  ## scale(PDI, center = TRUE, scale = TRUE)  0.0164  0.0267  0.6132  56  0.5422
 $\beta_2$  ## scale(cGDP, center = TRUE, scale = TRUE)  0.0358  0.0269  1.3314  56  0.1884
##                                     ci.lb      ci.ub
## intrcpt                           0.1093  0.1812  ***
## scale(PDI, center = TRUE, scale = TRUE) -0.0371  0.0698
## scale(cGDP, center = TRUE, scale = TRUE) -0.0180  0.0896
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Thank you.

